Endometriosis in Social Networks

Zita Berdefi (B7ZXSA)

Importance and relevance of the problem

Endometriosis affects roughly 10% of reproductive-age women worldwide (Organization 2021), yet diagnosis is often delayed for many years due to social stigma and the normalization of menstrual pain by both patients and physicians (Seear 2009; González-Espinoza and al. 2023). The condition is not only biomedical but also a social and informational phenomenon revealing structural inequalities in health awareness. Recent social media activism has increased visibility, reduced stigmatization, and encouraged earlier care-seeking (Adler, Wright, and Carter 2024). From a data-science and complexity perspective, the spread of awareness follows small-world diffusion patterns, where information and empathy travel through densely connected local clusters (Watts and Strogatz 1998). Thus, recognizing endometriosis is also a spatial-network problem shaped by social position, urban context, and online community structure.

State of research and approaches

While endometriosis-specific network studies are rare, related healthcare and social-network research can serve as a conceptual starting point. Small-world networks are characterized by short average path lengths and high clustering, which enable rapid diffusion of information and behavioral patterns (Watts and Strogatz 1998). This logic has been extended to economic and urban systems, where dense local structures and a few long-range connections facilitate innovation and knowledge flows (Barabási and Albert 1999; Fleming, King, and Juda 2007; Hausmann et al. 2014; Lengyel, Eriksson, et al. 2020). Within the framework of economic complexity, countries and industries grow by combining and reusing capabilities that are linked through networks of related products (Hidalgo and Hausmann 2009). Similar network-based mechanisms appear in health research, where spatial and social connections shape the diffusion of information and health outcomes. For instance, (Lengyel, Tóth, et al. 2024) show that antidepressant use is strongly correlated with spatial social network structures, highlighting how network position affects wellbeing. In a broader sense, disease and health complexity studies (Garas, Cimini, and Pammolli 2021) interpret comorbidities and health outcomes as parts of interconnected systems.

Research niche and question

Many studies have examined small-world and urban networks and their effects on health. Other research has focused on how endometriosis appears in social media and on the broader links between the economy and healthcare. However, these areas are usually studied separately. We still know little about how social networks—both online and offline—shape the spread of knowledge, the reduction of stigma, and the spatial pattern of diagnosis in chronic diseases such as endometriosis. Online communities and influencers may help information travel faster and make the condition more visible, but there is little empirical evidence for this. This study explores how the structure and dynamics of social networks shape the recognition and treatment of endometriosis. It examines whether higher network density, strong central actors, and greater online visibility lead to faster diffusion of awareness, lower stigma, and earlier diagnosis across social and spatial contexts.

Stakeholders and potential benefits

The main stakeholders are women affected by endometriosis and the healthcare professionals who diagnose and treat them. Greater public awareness can help patients recognize symptoms earlier and reduce the frustration that often arises from delayed or dismissive medical responses. For healthcare workers, understanding how information and attitudes spread through social networks may improve communication and diagnostic sensitivity. On a social level, the formation of smaller support clusters and online communities allows women to connect, share experiences, and feel less isolated. This can reduce stress—a known factor that worsens symptoms—and ultimately improve both wellbeing and healthcare outcomes.

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